Program : **Diploma in Printing Technology**	
Course Code : **3103**	Course Title: **Mechanism of Printing Machines-I**
Semester : **3**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester: **45**

Course Objectives:

- Getting the output through a printing machine is the most important operation for completing the print production.
- To provide a clear and sound knowledge in some of the major print production systems in Sheet fed and Web offset printing.
- This will impart the students to make judgment about the aspect of Offset printing, particularly the selection of a particular process to choose for a specific print production.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Definition of offset, Types of cylinders.		Basics of Printing Processes	2

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the characteristics of sheet fed offset printing machines.	15	Understanding
CO2	Demonstrate the printing process with sheet-fed offset machine.	14	Understanding
CO3	Demonstrate about web offset printing.	15	Understanding
CO4	Summarize web offset printing and troubleshooting in offset printing.	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain the characteristics of sheet fed offset printing machines.**		
M1.01	To classify offset printing machines,	3	Remembering
M1.02	To explain about feeding unit of sheet fed offset machine.	4	Understanding
M1.03	To describe about dampening unit of sheet fed offset machine.	4	Understanding
M1.04	To describe about inking unit of sheet fed offset machine.	4	Understanding
Contents: Definition of Lithography, Planography, Offset Printing. Classification of Offset printing machines - Sheet fed, web fed. Sheetfed Offset Machine: Introduction. Feeding unit: Types of Feeders - Friction feeders, Suction feeders, Types of Suction Feeders - Single sheet feeder and Stream Feeder. Pile board. Stoppers - front, side, back. Air blower nozzle. Suckers - lifting suckers, forwarding suckers. Flip arrangements - metal, brush. Feed board assemblies: introduction roller, drop down wheel, rubber wheel, brush wheels, endless tapes, ball arrangement. Modern feed board sheet forwarding. Registration unit: Front lay. Side lay - push type, pull type. Comparison between push type and pull type lays. Sheet detectors - no-sheet detector, multi sheet detectors, cross sheet detectors. Types of detectors (brief introduction only) - mechanical and photo electrical sheet detectors. Sheet insertion system/infeed unit: Swing arm, Rotary, Tumbler and Overfeed Grippers. Dampening system - types, construction and features. Contact and noncontact dampening system (Vibrator-type dampening system; Continuous-type dampening system; Indirect dampening solution application through the ink roller; Brush-type dampening system; Centrifugal dampening system; jet spray dampening system).Inking system- types, construction and features. (pyramid, multi roller) modern inking systems with motorized ink control. Digital inking system.			
CO2	**Demonstrate the printing process with sheet-fed offset machine.**		
M2.01	To understand the printing unit of a sheet fed offset machine and differentiate wet on wet and wet on dry printing.	2	Understanding

M2.02	To explain about Computer controlled offset machines and their features	2	Understanding
M2.03	To identify the sheet travel during perfecting and non-perfecting printing.	2	Understanding
M2.04	To describe about delivery unit of a sheetfed offset machine.	2	Understanding
M2.05	To describe about anti set off devices used in sheet fed offset printing	2	Understanding
M2.06	Differentiate the purpose of single-color sheet fed offset and multi-color offset machine.	2	Understanding
M2.07	To describe about make ready and pre-make ready operations.	2	Understanding
	Series Test I	1	

Contents:

Printing Unit: Parts of Plate cylinder, parts of Blanket cylinder, parts of Impression cylinder. Bearer contact and bearer clearance printing unit. Wet on wet printing and Wet on dry printing. Multicolor printing - with 4 color machines. Sheet travels in the perfecting sheet fed machine during the printing process. Computer controlled offset machines and their features - ink setting, sheet control, and registration control. Automatic blanket and impression cleaning systems. Automatic plate cleaning system. Delivery unit: Delivery Assist Devices, Suction Slow down Rollers, Blow downs, Wedges, Skeleton Wheels, Star Wheels. Anti-setoff devices - need of anti-setoff devices. Types of anti-setoff devices. Types of Sheet fed offset press - single color, multi-color. Perfecting presses. Sheet reversal systems. Anti-setoff devices - need of anti-setoff devices. Types of anti-setoff devices. Color sequence. Selection of color sequence - factors influencing color sequence. Color sequence used in the industry. Make ready and pre-make ready - definition, need, advantage, process, current industry practice, etc.

CO3	**Demonstrate about web offset printing.**		
M3.01	To define basic design features and working of web offset printing machine	3	Understanding
M3.02	To describe about infeed mechanism of web offset machine	4	Understanding
M3.03	To describe about dampening system of web offset machine	4	Understanding
M3.04	To describe about inking system of web offset machine	4	Understanding

Contents:

Web offset machine: Introduction, Overview. General terminology - web offset, printing couple, printing unit, perfecting, non-perfecting, inline, horizontal presses, vertical presses- I-C presses. Blanket to Blanket presses-Introduction, plate cylinder, Blanket cylinder, cylinder pressure &timing, Arithmetic oh packing. Packaging gauge, bench micrometer.

In feed section:

Single - roll stand, multiple roll stand, dancer roller, Lug air shaft, continuous roll feeding devices - Flying pasters - splicing sequence on flying pester Zero speed splicer - splicing sequence on a zero speed pester. Preparing a splice. Splice template, infeed operation.

Dampening system:

Types of dampening systems - levy flap dampening systems, continuous flow dampening systems, brush dampening using flick blades, Clare brush dampening systems, gross brush dampening systems. Alcohol in fountain solution. Continuous flow dampening systems - inker feed systems; Dahlgren dampening systems - Miehle - matic, Roland matic - Harris duotrol - Epic litho/dampener plate feed systems. Combination continuous - flow systems. Critical metering nip. Reverse slips nip - Smith dampening systems. Spray-bar dampening systems. Inking system - Introduction, functions of Inking system, construction of Inking system, roller setting methods, wash-up machines.

CO4	**Summarize web offset printing and troubleshooting in offset printing.**		
M4.01	To know about drying mechanism of heat set web offset machine	3	Understanding
M4.02	To know about chilling mechanism of heat set web offset machine	3	Understanding
M4.03	To explain the delivery mechanisms of web offset machine	3	Understanding
M4.04	To explain the inline finishing operations of web offset industry.	3	Understanding
M4.05	To identify the basic printing problems with offset printing	2	Understanding
	Series Test II	1	

Contents:

Dryers - introduction, function, setting of quick set ink, setting of heat set ink. Types of dryers, removal of solvent - ladder air from web, putting a controlled ripple in the web. Chill rolls - Introduction, function, types of roll system. The evolution of chill roll design, chill roll pumping. Folders - Introduction, folding principles, parts of folder, combination folder, ribbon folder, double - former folder, the mechanics of folding process of jaw fold, chopper fold mechanisms. Operation of collect cylinder, press folders, double former pre-folder, flow folders, insert folders. Inline finishing - Introduction, gluers, paster wheels, demonstrable pattern gluers, segmented gluers, envelope pattern gluers, backbone gluers. Pattern perforating and numbering units. Sheeters, variable rotary cutters. Troubleshooting - Troubles related to paper their remedies, Troubles related to ink and their remedies, Troubles related to dampening solution and their remedies, Troubles related to blanket and their remedies, Troubles related to printing machine and their remedies.

Text / Reference:

T/R	Book Title/Author
T1	Dejdas LP and Destree TM, **Sheet fed offset press operating**. GATF, USA
R1	Fonaid E Tood**Printing inks** PIRA international united Kingdom
R2	W.H. Bureau **What the printer should know about the paper**. GATF
R3	C S Misra, **Offset technology,** Anupam prakashan.
R4	Michael Bernard, John Peacock.,**Handbook of printing and production,** Springer Science & BusinessMedia
R5	Heigh. M. Speir, **Introduction in Printing Technology.**
R6	Adams J.M, Faux,D.D and Rieber L.J, **Printing Technology,** Delmar Publishers, NewYork
R7	Helmut Kipphan, **Hand book of print media,** Springer Science & Business Media.
R8	**GATF staff Solving sheet fed offset press problems,** GATF, USA.

Online resources:

Sl.No	Website Link
1	https://www.springer.com › Home › Printing & Publishing
2	https://www.heidelberg.com/in/en/index.jsp